

**PART-B**

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**INSTRUCTIONS**

1. Check that you have **10 printed pages**. The PART-B of the examination has **4 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No questions require any clarification from the instructor or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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|           |   |   |   |     |           |
|-----------|---|---|---|-----|-----------|
| Question: | 1 | 2 | 3 | 4   | Total     |
| Points:   | 6 | 7 | 5 | 2   | 20        |
| Score:    | 0 | 7 | 2 | 1+1 | 10+1 = 11 |

## QUESTIONS

1. (6 points) Let

$$S = \begin{pmatrix} S_{yy} & S_{yX}^T \\ S_{yX} & S_{XX} \end{pmatrix}$$

be the partitioned sample covariance matrix of  $Y_{1 \times 1}$  and  $X_{q \times 1}$  based on a centered (with respect to sample mean) sample of size  $n$ . Prove that the squared sample first canonical correlation  $r_1^2$  is identical to the coefficient of determination ( $R^2$ ) obtained from the multiple linear regression of  $Y$  on the vector  $X$ .

$$\text{Sol)} \quad S = \frac{1}{n-1} A_c^T A_c$$

$$Y = A[Y]_{1 \times 1}, \quad X = \begin{bmatrix} x_1 \\ \vdots \\ x_q \end{bmatrix}_{q \times 1}$$

$$S_{11} = S_{yy} = \frac{1}{n-1} [Y - \bar{y}] [Y - \bar{y}] = \frac{1}{n-1} (Y - \bar{y})_{1 \times 1}^T$$

$$S_{12} = S_{yx} = \frac{1}{n-1} \sum (x_i - \bar{x})(y - \bar{y}) = \begin{bmatrix} x_1 - \bar{x} \\ \vdots \\ x_q - \bar{x} \end{bmatrix}_{q \times 1} [Y - \bar{y}]_{1 \times 1} = \begin{bmatrix} (x_1 - \bar{x})(y - \bar{y}) \\ \vdots \\ (x_q - \bar{x})(y - \bar{y}) \end{bmatrix}_{1 \times 1}$$

$$S_{21} = S_{yx}^T = \begin{bmatrix} (x_1 - \bar{x})(y - \bar{y}) & \dots & (x_q - \bar{x})(y - \bar{y}) \end{bmatrix}$$

$$S_{xx} = \sum (x_i - \bar{x})^2 = \frac{1}{n-1} X_c^T X_c = \begin{bmatrix} x_1 - \bar{x} \\ \vdots \\ x_q - \bar{x} \end{bmatrix} \begin{bmatrix} x_1 - \bar{x} & \dots & x_q - \bar{x} \end{bmatrix} = \begin{bmatrix} x_1 - \bar{x} \\ \vdots \\ x_q - \bar{x} \end{bmatrix} \begin{bmatrix} x_1 - \bar{x} & \dots & x_q - \bar{x} \end{bmatrix}$$

$$S_{11} = S_{yy} = \frac{1}{n-1} (Y - \bar{y})^2 = \frac{1}{n-1} [(Y - \bar{y})^T]_{1 \times 1}$$

$$S_{21} = S_{yx} = \frac{1}{n-1} \sum (x_i - \bar{x})(y - \bar{y}) = \begin{bmatrix} (x_1 - \bar{x})(y - \bar{y}) \\ \vdots \\ (x_q - \bar{x})(y - \bar{y}) \end{bmatrix}_{q \times 1}$$

$$S_{12} = S_{yx}^T = \frac{1}{n-1} [(x_1 - \bar{x})(y - \bar{y}) + \dots + (x_q - \bar{x})(y - \bar{y})]_{1 \times q}$$

$$S_{22} = S_{xx} = \frac{1}{n-1} \sum (x_i - \bar{x})^2 = \frac{1}{n-1} \begin{bmatrix} x_1 - \bar{x} \\ \vdots \\ x_q - \bar{x} \end{bmatrix} \begin{bmatrix} x_1 - \bar{x} & \dots & x_q - \bar{x} \end{bmatrix} = \frac{1}{n-1} [(x_1 - \bar{x})^2 + \dots + (x_q - \bar{x})^2]$$

$$R^2 = 1 - \frac{SS_{\text{res}}/n-p-2}{SS_{\text{Tot}}/n-1}$$

$$M = \begin{bmatrix} -1 & -1 \\ S_{11} & S_{12} \end{bmatrix} \begin{bmatrix} -1 \\ -1 \end{bmatrix} \begin{bmatrix} S_{11}^{-1} & S_{12}^{-1} \end{bmatrix} \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

$$S_{22}^{-1} = \frac{1}{\sum (x_i - \bar{x})^2} \begin{bmatrix} \sum (x_i - \bar{x})^2 \end{bmatrix}$$

$$S_{11}^{-1} S_{12} = \frac{1}{(n-1)(Y - \bar{y})^2} \left[ (Y - \bar{y})^2 \times \sum (x_i - \bar{x})(y - \bar{y}) \right]$$

$$S_{22}^{-1} S_{21} = \frac{1}{\sum (x_i - \bar{x})^2} \left[ \sum (x_i - \bar{x})^2 \cdot \sum (x_i - \bar{x})(y - \bar{y}) \right]$$

$$\Rightarrow M = \frac{1}{(n-1)^2} \frac{(y-\bar{y})^2 (\sum (x_i - \bar{x})^2)}{(n-q-2)} \left[ \frac{(y-\bar{y})^2 \sum (x_i - \bar{x})^2}{(n-q-2)} \right]$$

$$= \left( \frac{1}{(n-1)^2} \frac{(y-\bar{y})^2 \sum (x_i - \bar{x})^2}{(n-q-2)} \right) \times 1$$

$$\lambda^2 \Rightarrow \lambda = \frac{n-1}{n-q-2}$$

$$S_{yy} = \frac{1}{n-1} (y-\bar{y})^2 \mathbf{I}_{1 \times 1}$$

$$S_{xx} = \frac{1}{n-q-2} (\sum (x_i - \bar{x})^2) \mathbf{I}_{1 \times 1}$$

$$S_{yx} = \frac{1}{(n-1)(n-q-2)} (\sum (x_i - \bar{x})) (y - \bar{y}) \mathbf{I}_{1 \times 1}$$

$$M = S_{11}^{-1} S_{12} + S_{22}^{-1} S_{21} = \frac{1}{(n-1)} \frac{1}{\left(\frac{1}{n-1}\right) (y-\bar{y})^2} \times \frac{1}{n-1} (y-\bar{y})^2 \mathbf{I}_{1 \times 1}$$

$$S_{22}^{-1} = \frac{1}{n-q-2} (\sum (x_i - \bar{x})^2) \mathbf{I}_{1 \times 1}$$

$$r_1^2 = \frac{\frac{1}{(n-1)(n-q-2)} (\sum (x_i - \bar{x})) (y - \bar{y})}{\frac{1}{(n-q-2)} (\sum (x_i - \bar{x})^2) \times \frac{1}{n-1} (y-\bar{y})^2} = \frac{\frac{1}{n-q-2} \times \sum (x_i - \bar{x})^2}{\frac{1}{n-1} (\sum (x_i - \bar{x})^2)}$$

$$r_1^2 = 1 - \frac{\frac{1}{n-q-2}}{\frac{1}{n-1}} \cdot \frac{\sum (x_i - \bar{x})^2}{SS_{\text{Total}}}$$

$$\Rightarrow R^2 = r_1^2 = R^2$$

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2. Suppose you are performing a factor analysis on a  $4 \times 4$  sample covariance matrix  $S$ . The eigenvalues of  $S$  are  $\lambda_1 = 0.8$ ,  $\lambda_2 = 2.1$ ,  $\lambda_3 = 4.2$ , and  $\lambda_4 = 0.2$ . The corresponding normalized eigenvectors are  $e_1, e_2, e_3$ , and  $e_4$ . You decide to extract  $m = 2$  factors using the principal component method.

(a) (2 points) Write down the explicit mathematical expression for the estimated factor loading matrix  $\tilde{L}$  in terms of the given eigenvalues and eigenvectors.

(b) (1 point) Let the first row of  $\tilde{L}$  be  $\tilde{l}_1^T = [0.8, -0.4]$ . Calculate the estimated communality  $\tilde{h}_1^2$  for the first variable.

(c) (4 points) Prove that the sum of all estimated communalities  $\sum_{i=1}^p \tilde{h}_i^2$  is exactly equal to the sum of the largest  $m = 2$  eigenvalues of  $S$ .

Sol)  $\lambda_3 > \lambda_2 > \lambda_1 > \lambda_4$ . Since only 2 factors to be extracted we take first 2 i.e.,  $\lambda_3, \lambda_2$ .

a)  $\Rightarrow \tilde{L} = [\sqrt{\lambda_3} e_3 \quad \sqrt{\lambda_2} e_2] = [\sqrt{4.2} e_3 \quad \sqrt{2.1} e_2] = [2.049 e_3 \quad 1.449 e_2]$ .

where  $e_3, e_2$  are  $4 \times 1$  matrices

b) Given 1<sup>st</sup> row of  $\tilde{L}^T = [0.8, -0.4]$

$$\Rightarrow \begin{bmatrix} 2.049 \\ 1.449 \end{bmatrix} \begin{bmatrix} e_{31} \\ e_{32} \\ e_{33} \\ e_{34} \end{bmatrix} + \begin{bmatrix} -1.449 \\ 1.449 \end{bmatrix} \begin{bmatrix} e_{21} \\ e_{22} \\ e_{23} \\ e_{24} \end{bmatrix} = \begin{bmatrix} \alpha \\ \gamma \\ a \\ c \end{bmatrix} \quad \text{--- (1)}$$

Given  $\alpha = 0.8, \beta = -0.4$

We know that  $\tilde{h}_i^2 = \sum_{j=1}^m l_{ij}^2 = l_{i1}^2 + l_{i2}^2$  (As  $m=2$  here)

Communality

$$\Rightarrow \tilde{h}_1^2 = l_{11}^2 + l_{12}^2 = (0.8)^2 + (-0.4)^2 = 0.64 + 0.16 = 0.8$$

$$\Rightarrow \boxed{\tilde{h}_1^2 = 0.8}$$

c)  $\sum \tilde{h}_i^2 = \sum_{i=1}^n \sum_{j=1}^m l_{ij}^2 = \text{Sum of all rows (sum of 1 row over all columns) of } \tilde{L}$ .

Let  $\tilde{L} = \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \\ a & b \\ c & d \end{bmatrix}$  (From (1))

$$\Rightarrow \sum \tilde{h}_i^2 = (\alpha^2 + \beta^2) + (\gamma^2 + \delta^2) + (a^2 + b^2) + (c^2 + d^2)$$

We know that  $L = [\sqrt{\lambda_3} e_3 \quad \sqrt{\lambda_2} e_2]$

$$\Rightarrow \sum \tilde{h}_i^2 = \lambda_3 \sum (e_{3i}^2) + \lambda_2 \sum (e_{2i}^2)$$

$$\Rightarrow \sum \tilde{h}_i^2 = \lambda_3 \sum (e_3^2) + \lambda_2 \sum (e_2^2)$$

$$\Rightarrow \tilde{L}^T L = [\alpha^2 + \beta^2 + \gamma^2 + \delta^2 + a^2 + b^2 + c^2 + d^2]$$

$$= [\lambda_2 e_2^2 + \lambda_3 e_3^2]$$

$$= [\lambda_2 + \lambda_3] \quad (\text{as } (e_2^2) = (e_3^2) = 1)$$

Here we get only ~~sum of max~~ largest  $m=2$  eigenvalues of  $S$  because we consider only the largest  $m'$  ~~max~~  $\lambda'$  values based on the principal component method.

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3. (5 points) Let  $X_1, X_2, \dots, X_n$  be a random sample of size  $n$  from a distribution with mean  $\mu \in \mathbb{R}$  and variance  $\mu^2$ . Let  $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ . Derive a variance-stabilizing transformation  $g(\bar{X}_n)$  such that the asymptotic variance of  $g(\bar{X}_n)$  does not depend on  $\mu$ .

Sol)  $\bar{X}_n = \frac{1}{n} \sum X_i$  let  $Y_n = \bar{X}_n$   
 Since mean of  $X_i$ 's =  $\mu$ .

By Delta Method:  $\sqrt{n}(\bar{X}_n - \mu) \rightarrow N(0, \mu^2)$   
 $\Rightarrow \sqrt{n}(g(\bar{X}_n) - g(\mu)) \rightarrow N(0, \frac{\mu^2}{n} (g'(\mu))^2)$

$\sqrt{n}(Y_n - \mu) \rightarrow N(0, \frac{\mu^2}{n})$   
 $\sqrt{n}(g(Y_n) - g(\mu)) \rightarrow N(0, \frac{\mu^2}{n} (g'(\mu))^2)$

$\Rightarrow (g'(\mu))^2 \propto \frac{1}{\mu^2} \Rightarrow g'(\mu) \propto \frac{1}{\mu} \Rightarrow g'(\mu) = \frac{c}{\mu} \rightarrow$  For  $\mu$  not to depend on  $\mu$ .  
 $\Rightarrow g(\mu) = c \ln \mu$

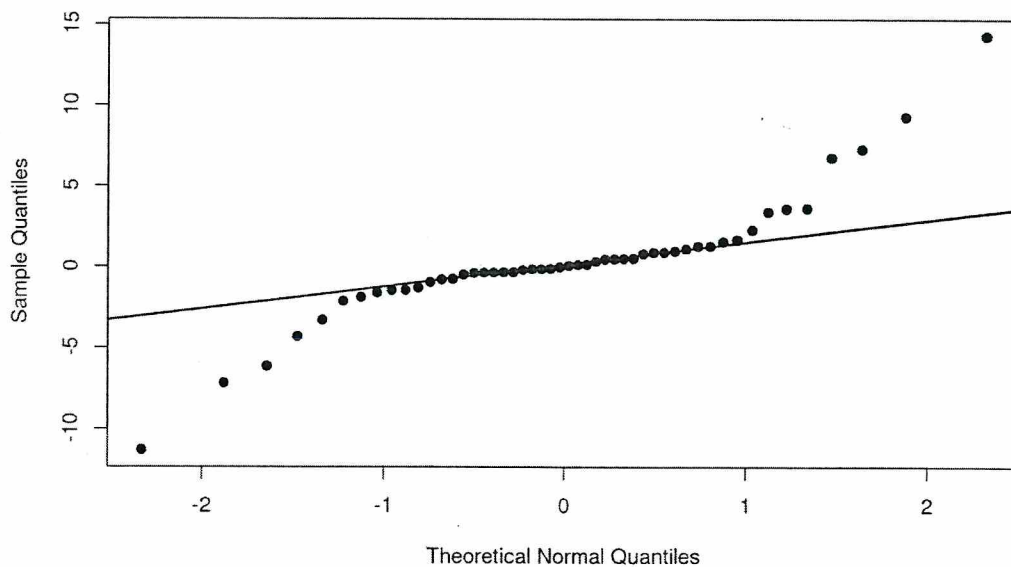
02

So  $g(Y_n) = c \ln Y_n \Rightarrow \boxed{g(\bar{x}) = c \ln \bar{x}}$

var of  $g(\bar{x})$

For  $\mu$  not to depend on  $\mu$ .

4. (2 points) The figure below displays a Normal Q-Q plot for a dataset of size  $n = 50$ . Interpret this plot to determine if the data follows a normal distribution. If the normality assumption is violated, specify which class of distributions would more accurately characterize the data.



- No this plot doesn't follow the ideal Normal distribution, since ~~at~~ the points near the tail deviate from the straight line behavior.
- This plot likely follows the ~~the~~ long tailed distribution, since ~~at~~ at the tails plot deviates too much.

what is it??

~~01~~

02

## ADDITIONAL PAGES FOR ANSWERS

The first, most obvious, benefit of the new system is that it will be more efficient and less costly than the old system. The second benefit is that it will be more secure and less vulnerable to attack. The third benefit is that it will be more flexible and easier to use. The fourth benefit is that it will be more reliable and less prone to error. The fifth benefit is that it will be more scalable and able to handle more data. The sixth benefit is that it will be more compatible and able to work with other systems. The seventh benefit is that it will be more user-friendly and easier to learn. The eighth benefit is that it will be more robust and able to handle more stress. The ninth benefit is that it will be more secure and less vulnerable to attack. The tenth benefit is that it will be more flexible and easier to use.





